SOIL EROSION AND CONSERVATION · NOTE ER05

Soil Loss Estimation — USLE Factors, Sediment Yield, MSLE, RUSLE and WEPP

Course topics: SWCE4.5 L4, SWCE-501 T8, SWCE-501 T11, SWCE-505 T10

Why estimate soil loss

Erosion reduces nutrients and moisture and so threatens productivity; on shallow, sloping soils severe erosion can destroy it completely. Off-site it causes sedimentation and pollution. Estimating soil loss lets us evaluate farming systems, plan control measures, and find the annual loss from a unit area under different rainfall, slope and other conditions.

Universal Soil Loss Equation (USLE)

$$A = R \cdot K \cdot (LS) \cdot C \cdot P$$

Symbol	Factor
A	Average annual soil loss (t/acre or t/ha)
R	Rainfall erosivity factor (erosion index, EI units)
K	Soil erodibility factor
LS	Topographic factor — slope length L and steepness S
C	Crop-management factor
P	Erosion-control practice factor

Scope: sheet and rill erosion on uplands. It excludes stream-bank erosion, snowmelt, wind erosion and deposition zones.

R — rainfall erosivity

R represents raindrop impact and runoff rate: the sum of the EI values of individual storms in the period, R = ΣEI. It is the main driver of water erosion and land degradation in India.

K — soil erodibility

- Soil loss per unit of erosivity (R or EI) from the **standard unit plot: 22.1 m long, 9% slope, continuous fallow**.
- K falls with better structure, coarser texture, more organic matter and more permeable lower horizons.

$$K = \text{Total adjusted soil loss} / \text{Total EI}$$

LS — topographic factor

$$L = (X/22.1)^{m}\ X = \text{slope length (m)}$$
$$S = 65.41 \cdot \sin^2\theta + 4.56 \cdot \sin\theta + 0.065\ \theta = \text{slope angle (from slope \%)}$$
$$LS = (X/22.1)^{m} \cdot (65.41 \cdot \sin^2\theta + 4.56 \cdot \sin\theta + 0.065)$$

Slope	Exponent m
> 5%	0.5
3–5%	0.4
1–3%	0.3
< 1%	0.2

Correction: the notes print the S equation with 4.58 but the combined LS equation with 4.56. The Wischmeier–Smith (1978) coefficient is 4.56. Check: at a 9% slope (sin θ = 0.0896) S = 0.525 + 0.409 + 0.065 ≈ 1.0, as it must be for the unit plot.

> **Illustration (not in the notes)**
> Slope length X = 44.2 m, slope 6% → θ = 3.43°, sin θ = 0.0599, m = 0.5
> L = (44.2/22.1)^0.5 = 1.414
> S = 65.41 × 0.00359 + 4.56 × 0.0599 + 0.065 = 0.573
> LS = 1.414 × 0.573 ≈ 0.81

C — crop management

C reflects crop cover and density, root growth and transpiration, crop sequence and productivity, tillage and residue management, and how erosive storms coincide with crop stages. C varies with crop, stage and place and is often found from runoff plots. The year is split into five crop-stage periods:

1. **Rough fallow** — summer ploughing to sowing.
2. **Seedbed** — seeding to 1 month.
3. **Establishment** — 1–2 months after seeding.
4. **Growing period** — establishment to harvest.
5. **Residue/stubble** — harvest to next ploughing.

P — conservation practice

P = soil loss under a practice ÷ soil loss under up-and-down-slope cultivation. Examples: up-and-down cultivation P = 1 (most hazardous); contour farming ≈ 0.68; strip cropping ≈ 0.51; terracing with contouring ≈ 0.38. P varies with slope range (tabulated in the textbook).

Uses of the USLE

- Predict average annual soil loss for a land-use pattern; guide the choice of crops and conservation practices; predict how management changes alter soil loss; estimate loss from non-agricultural land; supply data to planning agencies.

Sediment yield

$$Y = E \cdot (DR)\ Y = \text{sediment yield; } E = \text{gross erosion; } DR = \text{sediment delivery ratio}$$

Divide the watershed into sub-areas, apply the USLE to each and multiply by the delivery ratio.

MSLE, RUSLE and WEPP

Model	What changes
Modified Soil Loss Equation (MSLE)	A = R·K·(LS)·(VM). The **vegetation-management factor** VM (soil loss under vegetation ÷ loss from fallow) replaces C and P. VM = product of three sub-factors: canopy cover, ground cover (mulch and vegetation), and fine roots in bare soil
Revised USLE (RUSLE)	Overcomes USLE limitations with expanded data sets, corrections for litter biomass, canopy, plant height and ground cover, and computer-based calculation. Applies to agricultural, grazing and forest land with more flexibility and accuracy
WEPP (Water Erosion Prediction Project)	A process-based, time-step watershed model of soil, management, biomass, canopy and ground cover; estimates erosion, sediment transport and deposition within watersheds

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. Wischmeier, W. H. and Smith, D. D. (1978), Predicting Rainfall Erosion Losses, USDA Agriculture Handbook 537. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. The standard USLE unit plot is

(a) 10 m long, 5% slope, cropped
(b) 22.1 m long, 9% slope, continuous fallow
(c) 22.1 m long, 1% slope, grass
(d) 100 m long, 9% slope, fallow

Q2. In the USLE, the slope-length exponent m for slopes above 5% is

(a) 0.2
(b) 0.3
(c) 0.4
(d) 0.5

Q3. P = 1 corresponds to

(a) contour farming
(b) terracing
(c) up-and-down slope cultivation
(d) strip cropping

Q4. Approximate P for strip cropping is

(a) 0.38
(b) 0.51
(c) 0.68
(d) 1.0

Q5. The USLE does NOT estimate

(a) sheet erosion
(b) rill erosion
(c) stream-bank erosion
(d) upland soil loss

Q6. Sediment yield = gross erosion ×

(a) runoff coefficient
(b) delivery ratio
(c) C factor
(d) trap efficiency

Q7. In the MSLE, C and P are replaced by

(a) LS
(b) VM
(c) K
(d) R

Q8. R in the USLE equals

(a) ΣEI of storms
(b) mean annual rainfall
(c) maximum intensity
(d) runoff depth

Answers

Q1 — (b) 22.1 m long, 9% slope, continuous fallow

Q2 — (d) 0.5
Q3 — (c) up-and-down slope cultivation
Q4 — (b) 0.51
Q5 — (c) stream-bank erosion
Q6 — (b) delivery ratio
Q7 — (b) VM
Q8 — (a) ΣEI of storms
